ASCII MASTER FLOWCHART

Multi-Hazard Early Warning and Disaster Technology

12-panel revision companion | GS-III Disaster Management

ASCII MASTER FLOW — PANEL 1/12: End-to-end people-centred chain

```
RISK KNOWLEDGE -> MONITOR / FORECAST -> AUTHORITATIVE WARNING
-> MULTI-CHANNEL DISSEMINATION -> COMPREHENSION -> EARLY ACTION
-> OUTCOME + FEEDBACK -> UPDATED RISK KNOWLEDGE
```

ASCII MASTER FLOW — PANEL 2/12: Risk-knowledge and GIS stack

```
HAZARD + EXPOSURE + VULNERABILITY - CAPACITY = DISASTER RISK
GIS overlays people, assets, lifelines, routes and shelters.
Remote sensing observes; GIS integrates; institutions decide; field teams act.
```

ASCII MASTER FLOW — PANEL 3/12: Forecast-warning-order firewall

```
OBSERVATION -> FORECAST / ASSESSMENT -> AUTHORISED WARNING -> ACTION ORDER
Nowcasting: conventionally present to ~6 h; IMD commonly ~3 h products.
Five-to-thirty minutes is a narrow extrapolation example, not a universal rule.
```

ASCII MASTER FLOW — PANEL 4/12: Hazard and lead-time matching

```
EARTHQUAKE -> no deterministic prediction; rapid post-event parameters
TSUNAMI -> under-10-min earthquake detection; longer public-action chain
URBAN RAINFALL -> qualified 3-6 h source planning example
CYCLONE -> longer tracking window; action still depends on preparedness
```

ASCII MASTER FLOW — PANEL 5/12: CAP-SACHET dissemination

```
COMPETENT AGENCY -> ONE STRUCTURED CAP ALERT -> NDMA SACHET
-> SMS / CELL BROADCAST / APP / WEB / RADIO / TV / SIREN
-> ACCESSIBLE LOCAL RELAY -> RECEIPT TEST
CAP solves format interoperability, not mandate or readiness.
```

ASCII MASTER FLOW — PANEL 6/12: Warning-to-action test

```
DELIVERED? -> RECEIVED? -> UNDERSTOOD? -> TRUSTED? -> ACTION FEASIBLE?
ROUTE + TRANSPORT + SHELTER + SUPPLIES + ASSISTANCE AVAILABLE?
Success is the at-risk person protected, not the message sent.
```

ASCII MASTER FLOW — PANEL 7/12: Agency ownership

```
IMD -> WEATHER / CYCLONE | CWC -> RIVER FLOOD
INCOIS / ITEWC -> TSUNAMI | NCS / MoES -> EARTHQUAKE PARAMETERS
ISRO / NRSC -> EARTH OBSERVATION / GIS
NDMA + STATE / LOCAL -> INTEGRATION AND PROTECTIVE ACTION
```

ASCII MASTER FLOW — PANEL 8/12: Public applications

```
MAUSAM / MEGHDOOT -> IMD | DAMINI -> IITM PUNE
BHOOKAMP -> NCS / MoES | UMANG -> MeitY platform
SACHET -> NDMA integrated alert portal
APP EXISTENCE != UNIVERSAL REACH OR PROTECTIVE OUTCOME
```

ASCII MASTER FLOW — PANEL 9/12: Phailin and impact-based warning

```
SATELLITE + RADAR + MODELS -> ACCURATE TRACK / INTENSITY FORECAST
-> AUTHORISED IMPACT MESSAGE -> MASS PREPAREDNESS / EVACUATION
Use "forecasting and tracking", not deterministic "exact prediction".
Impact message names consequence and action, not only wind or rainfall.
```

ASCII MASTER FLOW — PANEL 10/12: Resilient architecture

```
DIVERSE SENSORS + COMMON STANDARDS + BACKUP POWER / NETWORK
+ MAINTENANCE + CYBERSECURITY + MANUAL LOCAL PATH
One failure should degrade the chain, not silence it.
```

ASCII MASTER FLOW — PANEL 11/12: Emerging-technology safeguards

```
DRONE / AI / CROWD REPORT -> PROVENANCE + HUMAN VALIDATION
-> UNCERTAINTY + NECESSITY + DATA MINIMISATION + ACCESS CONTROL
-> INCLUSIVE NON-DIGITAL ALTERNATIVE
TECHNICAL INPUT != VERIFIED OUTCOME
```

ASCII MASTER FLOW — PANEL 12/12: EW4All answer spine

```
TARGET 2027
UNDRR -> RISK KNOWLEDGE | WMO -> DETECTION / FORECASTING
ITU -> DISSEMINATION | IFRC -> PREPAREDNESS / RESPONSE
Assess each pillar; do not present the target as achieved coverage.
```